

ENGRD 2700

Homework 1

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1.

(a) Compute the sample mean, sample median, and sample standard deviation for each column of the dataset.

For the first column,

$$x_1 = \{10, 8, 13, 9, 11, 14, 6, 4, 12, 7, 5\}, \quad n = 11.$$

The sample mean is

$$\bar{x}_1 = \frac{\sum_{i=1}^n x_i}{n} = \frac{99}{11} = 9.$$

To find the sample median, first order the observations:

$$\{4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14\}.$$

Since $n = 11$ is odd,

$$\tilde{x}_1 = x_{(\frac{n+1}{2})} = x_{(6)} = 9.$$

The variance is

$$\sigma_{x_1}^2 = \frac{\sum_{i=1}^n (x_i - \bar{x}_1)^2}{n - 1}.$$

Substituting the observations,

$$\begin{aligned} \sigma_{x_1}^2 &= \frac{(10 - 9)^2 + (8 - 9)^2 + (13 - 9)^2 + (9 - 9)^2 + (11 - 9)^2}{11 - 1} \\ &\quad + \frac{(14 - 9)^2 + (6 - 9)^2 + (4 - 9)^2 + (12 - 9)^2 + (7 - 9)^2 + (5 - 9)^2}{11 - 1} \\ &= 11. \end{aligned}$$

Therefore, the standard deviation is

$$\sigma_{x_1} = \sqrt{\sigma_{x_1}^2} = \sqrt{11} \approx 3.317.$$

The same calculations were performed for the remaining columns, resulting in the statistics below.

Column	Mean	Median	Variance	Std. Dev.
x_1	9.00	9.00	11.000	3.317
y_1	7.50	7.58	4.127	2.032
x_2	9.00	9.00	11.000	3.317
y_2	7.50	8.14	4.128	2.032
x_3	9.00	9.00	11.000	3.317
y_3	7.50	7.11	4.123	2.030
x_4	9.00	8.00	11.000	3.317
y_4	7.50	7.04	4.123	2.031

(b) Based solely upon the summary statistics you computed in part (a), how do the four datasets compare?

All summary statistics appear similar across the four datasets, with the same sample means and sample standard deviations. The sample medians are also similar, except for the y -values and x_4 , which have varying medians.

(c) Construct scatterplots for each of the four datasets.

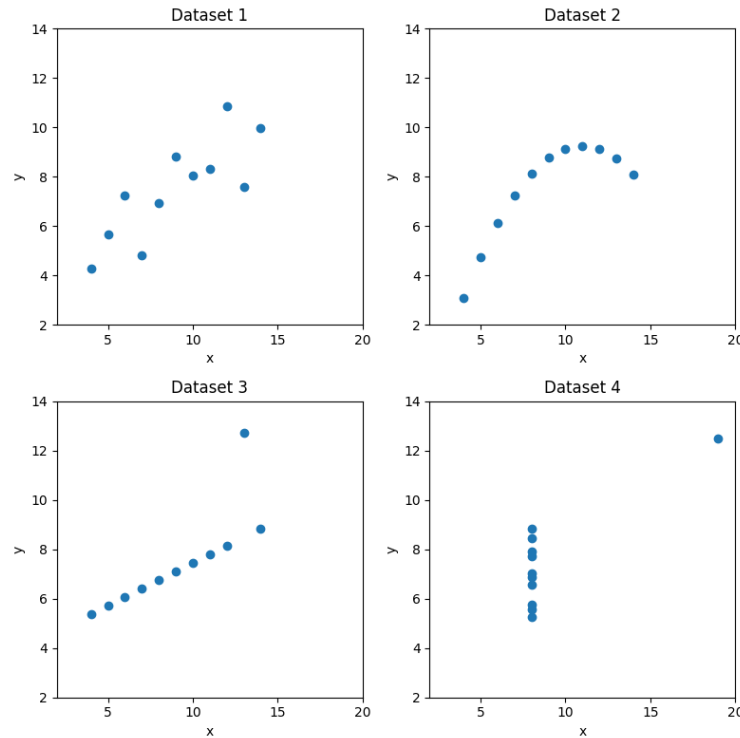


Figure 1: Scatterplots of the four datasets (x_1, y_1) , (x_2, y_2) , (x_3, y_3) , and (x_4, y_4) . [1]

(d) Based solely upon the plots generated in part (c), how do the four datasets compare?

The four datasets appear structurally different, with Dataset 1 having a roughly linear relationship, Dataset 2 showing a nonlinear curved relationship, Dataset 3 showing an extremely linear

relationship with one outlier, and Dataset 4 having the same x -values with one outlier far to the right.

(e) What is the moral of the story?

The four datasets looked extremely similar based on their summary statistics, but the scatterplots showed extremely dissimilar relationships. The moral is that we should not only use numerical summaries, we should also use visualizations to have a more holistic view of the data.

2.

(a) Provide a histogram of the per-county percentage of residents who speak a foreign language at home during 2006–2010.

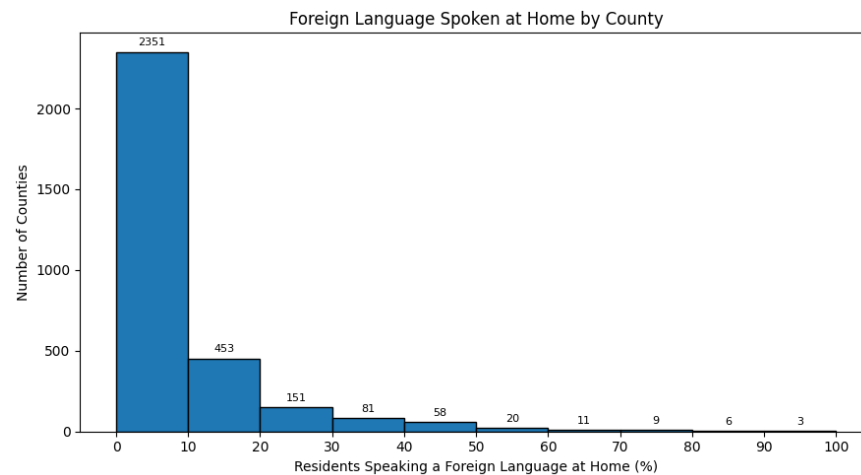


Figure 2: Distribution of the percentage of residents who speak a foreign language at home across U.S. counties. [2]

(b) What was the median per-county amount of federal spending in 2009?

The median per-county amount of federal spending in 2009 was

\$214,994 [3]

(c) Create a scatter plot of the percentage of residents below the poverty level (y-axis) versus the percentage of the population with a bachelors degree. Comment on what you see.

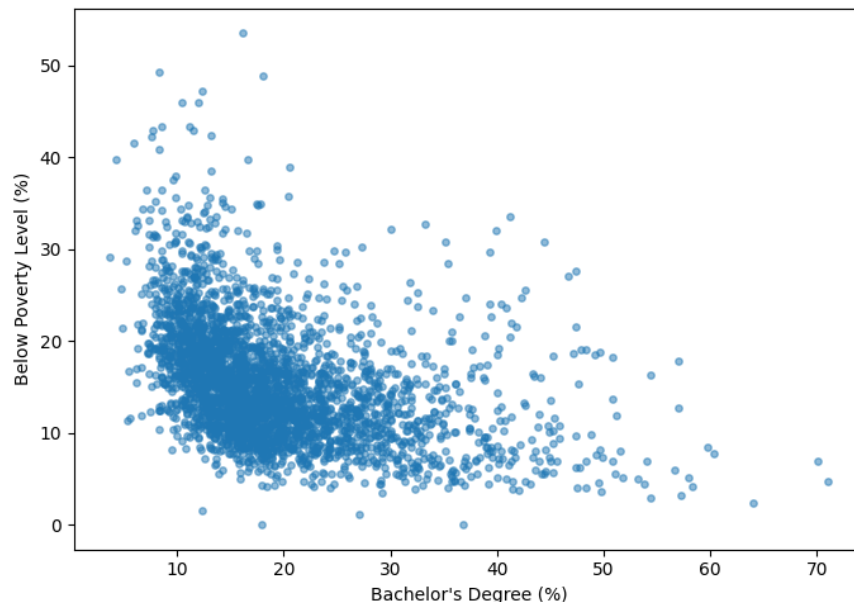


Figure 3: Percentage of residents below the poverty level versus the percentage of residents with a bachelor's degree. [4]

The scatterplot shows a clear negative relationship between bachelor's degree percentage and poverty percentage in counties, with counties with a higher percentage of bachelor's degree holders tending to have lower poverty percentages. However, there is still considerable variation, especially among counties with lower bachelor's degree percentages.

(d) What fraction of counties have a population whose percentage under the age of 18 is above 30%?

There are 114 counties out of 3,143 with more than 30% of their population under the age of 18. Therefore,

$$\frac{114}{3143} \approx 0.0363,$$

or approximately 3.63% of counties. [5]

3.

(a) What is the new mean house price?

The original set has 24 houses with a mean price of \$500,000. Since

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n},$$

the total price of the original 24 houses is

$$\sum_{i=1}^{24} x_i = 24(500,000) = 12,000,000.$$

After adding the new \$700,000 house, the total becomes

$$12,000,000 + 700,000 = 12,700,000.$$

There are now 25 houses, so the new mean is

$$\bar{x} = \frac{12,700,000}{25} = \boxed{\$508,000}.$$

(b) What is the new standard deviation?

The original standard deviation is

$$\sigma = 30,000.$$

Squaring both sides gives the original variance:

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}.$$

Substituting $n = 24$, $\bar{x} = 500,000$, and $\sigma = 30,000$,

$$(30,000)^2 = \frac{\sum_{i=1}^{24} (x_i - 500,000)^2}{23}.$$

Multiplying both sides by 23,

$$\sum_{i=1}^{24} (x_i - 500,000)^2 = 23(30,000)^2 = 20,700,000,000.$$

This is the total squared spread of the original 24 houses around the old mean of \$500,000.

From part (a), the new mean is \$508,000. The mean has shifted by

$$508,000 - 500,000 = 8,000.$$

Because the 24 original houses are now measured relative to this new center, the total squared spread increases by

$$24(8,000)^2 = 1,536,000,000.$$

So the original 24 houses now contribute

$$20,700,000,000 + 1,536,000,000 = 22,236,000,000$$

of total squared spread around the new mean.

The new \$700,000 house spread is

$$700,000 - 508,000 = 192,000$$

away from the new mean, so its squared spread is

$$(192,000)^2 = 36,864,000,000.$$

Therefore, the new total squared spread is

$$22,236,000,000 + 36,864,000,000 = 59,100,000,000.$$

As there are now 25 houses, the new variance is

$$\sigma^2 = \frac{59,100,000,000}{25 - 1} = 2,462,500,000.$$

And the new standard deviation is

$$\sigma = \sqrt{2,462,500,000} \approx \boxed{\$49,624}.$$

(c) Does the median increase, decrease, or stay the same after the new house is built? Or can no conclusion be made? Explain.

The new median cannot be determined exactly. The original dataset has 24 houses, so the median is based on the 12th and 13th ordered values,

$$\frac{x_{(12)} + x_{(13)}}{2} = 440,000.$$

After adding the new house, there are 25 houses, so the new median is $x_{(13)}$. Since the individual values of $x_{(12)}$ and $x_{(13)}$ are unknown, the exact new median cannot be determined. However, because the new \$700,000 house is above the original median, the median can only stay the same or increase. It cannot decrease.

4.

(a) Suppose that you modify these data by adding a constant c to each observation in the sample, and then multiplying by another constant k , to obtain $y_i = (x_i + c)k$. What are $\bar{y}$ and s_y^2 , the sample mean and variance of the modified data? Justify your answer mathematically using the definition of the sample mean and variance.

The sample mean is defined as

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}.$$

Adding c to every observation shifts the mean by c , so the mean becomes

$$\bar{x} + c.$$

Multiplying every observation by k then multiplies the mean by k . Therefore,

$$\boxed{\bar{y} = k(\bar{x} + c)}.$$

The sample variance is defined as

$$s_x^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}.$$

Adding c to every observation also adds c to the mean. Therefore, the deviation of each observation from the mean does not change:

$$(x_i + c) - (\bar{x} + c) = x_i - \bar{x}.$$

Multiplying every observation by k multiplies each deviation from the mean by k :

$$k(x_i - \bar{x}).$$

Since variance uses squared deviations, the new squared deviation is

$$(k(x_i - \bar{x}))^2 = k^2(x_i - \bar{x})^2.$$

Therefore, every squared deviation is multiplied by k^2 , while $n - 1$ remains unchanged. Thus,

$$\boxed{s_y^2 = k^2 s_x^2}.$$

(b) Finally, suppose we built another modified data sample $z_1, \dots, z_n$, where $z_i = \frac{x_i - \bar{x}}{s_x}$, where s_x is the sample standard deviation of the x -data. This procedure standardizes the original data. What are $\bar{z}$ and s_z^2 ? Justify your answer.

The sample mean of the standardized data is

$$\bar{z} = \frac{\sum_{i=1}^n z_i}{n}.$$

Substituting $z_i = \frac{x_i - \bar{x}}{s_x}$,

$$\bar{z} = \frac{1}{n} \sum_{i=1}^n \frac{x_i - \bar{x}}{s_x}.$$

Since s_x is the same for every observation,

$$\bar{z} = \frac{1}{s_x} \left(\frac{\sum_{i=1}^n (x_i - \bar{x})}{n} \right).$$

Since the deviations from the mean sum to zero,

$$\sum_{i=1}^n (x_i - \bar{x}) = 0.$$

Therefore,

$$\bar{z} = \frac{1}{s_x} \left(\frac{0}{n} \right) = \boxed{0}.$$

The sample variance of the standardized data is

$$s_z^2 = \frac{\sum_{i=1}^n (z_i - \bar{z})^2}{n - 1}.$$

Since $\bar{z} = 0$,

$$s_z^2 = \frac{\sum_{i=1}^n z_i^2}{n - 1}.$$

Substituting $z_i = \frac{x_i - \bar{x}}{s_x}$,

$$s_z^2 = \frac{\sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right)^2}{n - 1}.$$

Squaring the fraction gives

$$s_z^2 = \frac{1}{s_x^2} \left(\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1} \right).$$

The term in parentheses is the original sample variance s_x^2 . Therefore,

$$s_z^2 = \frac{1}{s_x^2} (s_x^2) = \boxed{1}.$$

Appendix

[1] Question 1(c) — Scatterplot Generation

```
1 import matplotlib.pyplot as plt
2
3 x1 = [10,8,13,9,11,14,6,4,12,7,5]
4 y1 = [8.04,6.95,7.58,8.81,8.33,9.96,7.24,4.26,10.84,4.82,5.68]
5
6 x2 = [10,8,13,9,11,14,6,4,12,7,5]
7 y2 = [9.14,8.14,8.74,8.77,9.26,8.10,6.13,3.10,9.13,7.26,4.74]
8
9 x3 = [10,8,13,9,11,14,6,4,12,7,5]
10 y3 = [7.46,6.77,12.74,7.11,7.81,8.84,6.08,5.39,8.15,6.42,5.73]
11
12 x4 = [8,8,8,8,8,8,19,8,8,8]
13 y4 = [6.58,5.76,7.71,8.84,8.47,7.04,5.25,12.50,5.56,7.91,6.89]
14
15 fig, axs = plt.subplots(2, 2, figsize=(8, 8))
16
17 datasets = [
18     (x1, y1, "Dataset 1"),
19     (x2, y2, "Dataset 2"),
20     (x3, y3, "Dataset 3"),
21     (x4, y4, "Dataset 4")
22 ]
23
24 for ax, (x, y, title) in zip(axs.flat, datasets):
25     ax.scatter(x, y)
26     ax.set_title(title)
27     ax.set_xlabel("x")
28     ax.set_ylabel("y")
29     ax.set_xlim(2, 20)
30     ax.set_ylim(2, 14)
31
32 plt.tight_layout()
33
34 plt.savefig(
35     "hw1-1c-scatterplots.png",
36     dpi=300,
37     bbox_inches="tight"
38 )
39
40 plt.show()
```

[2] Question 2(a) — Histogram

```
1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 df = pd.read_csv("CountyData.csv")
5
6 foreign_language = df["foreign_spoken_at_home"]
7
8 bins = range(0, 101, 10)
9
10 fig, ax = plt.subplots(figsize=(9, 5))
11
12 counts, _, bars = ax.hist(
13     foreign_language,
14     bins=bins,
15     edgecolor="black"
16 )
17
18 ax.bar_label(
```

```

19     bars,
20     labels=[str(int(x)) if x > 0 else "" for x in counts],
21     padding=3,
22     fontsize=8
23 )
24
25 ax.set_xlabel(
26     "Residents Speaking a Foreign Language at Home (%)"
27 )
28
29 ax.set_ylabel("Number of Counties")
30
31 ax.set_xticks(range(0, 101, 10))
32
33 plt.tight_layout()
34
35 plt.savefig(
36     "hw1-2a-histogram.png",
37     dpi=300,
38     bbox_inches="tight"
39 )
40
41 plt.show()

```

[3] Question 2(b) — Median Federal Spending

```

1 import pandas as pd
2
3 df = pd.read_csv("CountyData.csv")
4
5 median_fed_spending = df["fed_spending"].median()
6
7 print(median_fed_spending)

```

[4] Question 2(c) — Scatterplot

```

1 import pandas as pd
2 import matplotlib.pyplot as plt
3
4 df = pd.read_csv("CountyData.csv")
5
6 fig, ax = plt.subplots(figsize=(7, 5))
7
8 ax.scatter(
9     df["bachelors"],
10    df["poverty"],
11    alpha=0.5,
12    s=15
13 )
14
15 ax.set_xlabel("Bachelor's Degree (%)")
16 ax.set_ylabel("Below Poverty Level (%)")
17
18 plt.tight_layout()
19
20 plt.savefig(
21     "hw1-2c-scatterplot.png",
22     dpi=300,
23     bbox_inches="tight"
24 )
25
26 plt.show()

```

[5] Question 2(d) — Fraction of Counties

```
1 import pandas as pd
2
3 df = pd.read_csv("CountyData.csv")
4
5 count_above_30 = (df["age_under_18"] > 30).sum()
6 total_counties = len(df)
7
8 fraction = count_above_30 / total_counties
9
10 print(count_above_30)
11 print(total_counties)
12 print(fraction)
```